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Unit – 1 : Session – 11 : SLO – 2 | EX-NOR gate

■ II. Fill in the Blanks

1. same (equal)
2. $Y = (A \oplus B) = AB + A\bar{B}$
3. $\oplus$ (EX-OR with inversion bubble)
4. equality (error/parity)
5. equality

■ I. Match the Following

1. EX-NOR Logic Expression — b) $Y = (A \oplus B)$
2. Output Condition — a) 1 when inputs are equal
3. Symbol Used — c) $\oplus$ or $\oplus$
4. Truth Table Output for (1,0) — d) 0
5. Practical Application — e) Digital comparators and equality checking

■ III. Activity / Describe and Perform

- Logic function: EX-NOR is the complement of EX-OR — output 1 when both inputs are EQUAL. Boolean expression: $Y = (A \oplus B) = AB + A\bar{B}$.
- Symbol: EX-OR symbol with an inversion bubble at the output.
- Truth table: 00→1, 01→0, 10→0, 11→1.
- Practical application: digital comparators — one EX-NOR per bit; all bits equal → all outputs 1 → AND them to detect equality of two numbers. Verifying all four input combinations experimentally reproduces the truth table.